HUMZA KHAN

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EDUCATION

Northwestern University **Expected May 2026**

M.S. *Statistics and Data Science (B.S./M.S. Program)*

B.S. *Industrial Engineering and Management Sciences; Economics* GPA: 3.98/4.00

Relevant Coursework: Probability (grader and student), Statistics, Data Science, Applied Linear Algebra, Microeconomics, Macroeconomics, Data Management and Processing, Foundations of Optimization, Econometrics

Awards and Societies: Dean's List (6x), Tau Beta Pi

WORK EXPERIENCE

Fairmont Consulting Group, Summer Analyst **June 2025 – August 2025**

- Conducted analyses of DoD and other federal spending data to map market size, key suppliers, and emerging technologies for a variety of M&A and strategy engagements for defense and government services clients
- Leveraged Python and Excel to gather, process, and visualize large datasets to create actionable insights for client and internal leadership decisions
- Performed competitive landscape and capability-gap assessments, synthesizing open-source intelligence with proprietary research to spotlight high-value acquisition targets and growth adjacencies

P33 Chicago, Deep Tech Intern **June 2024 – August 2024**

- Researched the national and regional quantum ecosystem focusing on technology commercialization and models designed to accelerate the development of quantum, drawing on historical evidence from related hard-tech industries
- Developed regional quantum and advanced materials industry maps using large-scale open-source databases such as OpenAlex and the U.S. Patent Office
- Built a live grants dashboard for all funding opportunities related to CHIPS, IRA and IIJA to harmonize various tracking sources

U.S. Department of State, Technology Policy Intern **September 2023 – May 2024**

- Mapped global semiconductor R&D capabilities, encompassing basic research, physical infrastructure, applications, and end users, with a focus on geopolitical, innovation, and economic implications.
- Employed the US Patent Office API and Python to generate and visualize semiconductor ecosystem analytics at the country and institutional level from 250,000+ IEEE publications and 1.2 million patents

Haylon Technologies, Student Consultant **September 2022 – May 2023**

- Conducted market research on a variety of sectors including energy, batteries, and mining
- Researched product analogs and vetted potential investors to provide strategy and funding recommendations
- Developed SPICE and industry-standard PCB software models of battery components and cells, including a lithium-ion battery to within 5% accuracy of all characteristics, custom-modeled components, and overall cell integration

PROJECTS & PUBLICATIONS (available on personal website)

Scrambling for Power: The Impact of Sudden Reactor Shutdowns on Short-Run Electricity Prices; *ECON 383 - Applied Econometrics*: This study examines the impact of instantaneous nuclear reactor faults (scrams) on electricity prices. By implementing a stacked regression discontinuity in time (RDiT) design across 13 reactors, I find a causal effect of $26.18/MWh per lost GW of nuclear capacity.

Power Up; *Stanford Economic Review*: This commentary examines regulatory and policy-driven challenges facing U.S. energy infrastructure, with targeted policy reforms aimed at strengthening grid resilience and accelerating modernization.

International Semiconductor R&D Capabilities Mapping; *U.S. Department of State*: This interactive dashboard and report provides in-depth analyses of the global semiconductor ecosystem including specific facilities, research areas and international collaborations.

SKILLS & CERTIFICATES

Data Analysis and Econometrics: Python (Udemy Certification), R, SQL, Tableau, Excel, PowerPoint

Engineering Software: LTSpice, Altium, AMPL, Gurobi